

RAÚL JULIÁN REVELO TOBAR

Materials Scientist | Ph.D. in Materials Science & Engineering

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RESEARCH INTERESTS

Materials Scientist with over 10 years of research experience specializing in glass science, glass-ceramic systems, and functional silicate materials. My expertise lies in the nucleation and crystallization kinetics of oxide glass systems, with a focus on controlled nanocrystal precipitation, multiscale microstructural characterization, and structure–property relationships.

RESEARCH INTERESTS

- Development of Low Dielectric Constant (Low-k) and controlled CTE glass-ceramics for 5G/6G infrastructure and microelectronic packaging.
- Utilizing Machine Learning-aided formulation to predict and optimize high-refractive index glasses and functional glass-ceramics.
- Synthesis of Ultra-High Strength Transparent Glass-Ceramics for aerospace, defense, and high-durability electronic displays.

EDUCATION

Ph.D. in Materials Science & Engineering

January 2019

University of São Paulo (USP), São Carlos, Brazil

Thesis: *Fabrication and Characterization of Dielectric Glass-Ceramic Substrates Based on Diopside Formed by Tape Casting*

Advisor: Prof. Dr. Eduardo Bellini Ferreira

M.Sc. in Materials Science & Engineering

September 2014

University of São Paulo (USP), São Carlos, Brazil

Thesis: *Reformulation of Ceramic Glazes Using CRT Glass*

Advisor: Prof. Dr. Eduardo Bellini Ferreira

B.Sc. in Materials Engineering

December 2009

University of Antioquia, Medellín, Colombia

RESEARCH EXPERIENCE

Postdoctoral Researcher

June 2023 – November 2024

Vitreous Materials Laboratory (LaMaV), Federal University of São Carlos (UFSCar), Brazil

Advisor: Prof. Dr. Edgar D. Zanotto | Funded by: AGC Inc. (Japan)

- Led optimization of LAS ($\text{Li}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot n\text{SiO}_2$) glass-ceramics for an international industry-academia project with AGC Inc. (Japan), developing treatment routes for controlled ZrO_2 nanocrystal precipitation.
- Investigated early and advanced crystallization stages in glasses and supercooled liquids; engineered nucleation conditions to achieve homogeneous nanostructures linking scientific findings with industrial manufacturing targets.
- Performed comprehensive nano and microstructural characterization (DSC, SEM, STEM, XRD, Raman spectroscopy) to guarantee material stability, reproducibility, and performance.

Postdoctoral Researcher

March 2020 – April 2022

Research Group on Functional Materials and Catalysis (GIMFC), University of Nariño, Pasto, Colombia

Advisor: Prof. Dr. Luis Alejandro Galeano

- Evaluated glass-ceramic photocatalysts for water purification via visible-light Fenton-like processes; designed materials with controlled nanostructured phase crystallization.

- Characterized surface-analyte interactions with complex organic pollutants (Ibuprofen, Natural Organic Matter) in liquid matrices using HPLC, UV-VIS, and DOC analysis at ppm/ppb concentrations.

INDUSTRY & ENTREPRENEURSHIP

Co-Founder & Operations Manager

January 2018 – April 2023

GAIA GreenTech — PIPE/FAPESP Program, São Carlos, Brazil

- Built and led for 5 years a circular economy startup focused on smart management of electronic waste.
- Developed FAPESP-funded systems for e-waste traceability using deep learning and blockchain.
- Led all design, implementation, validation phases; presented results to funding agencies.

Technical & Quality Manager

April 2010 – July 2012

CENTREK, Colombia

- Managed laboratory quality control, equipment maintenance, calibration protocols, and technical operations.

TECHNICAL SKILLS & INSTRUMENTATION

- **Structural Analysis:** STEM, TEM, SEM, XRD, XRF, Raman Spectroscopy, UV-Vis.
- **Thermal Analysis:** DSC, DTA, TGA, Dilatometry.
- **Advanced Processing:** Aqueous Tape Casting (thin substrates), Sol-Gel, Pechini Method, Controlled Nanocrystal Precipitation.
- **Optical/Functional:** Photoluminescence (Rare-earth probes), BET Surface Area, DOC/TOC.
- **Digital Tools:** Deep Learning (DeepLearning.AI), Python.

PEER-REVIEWED PUBLICATIONS

1. Submitted CERI-D-26-04787; Revelo R.J., Gomes Machado Filho S., Fortulan C.A. & Ferreira E.B. (2026). **Preparation and Characterization of Diopside Glass-Ceramic Substrates via Aqueous Tape Casting.** *Ceramics International*.
2. In preparation: Revelo R.J., Souza M.L., Pena R.B., Merízio L.G., Yoshida S. & Zanotto E.D. (2026). **Early Stages of ZrO₂-induced Nucleation in Lithium Aluminosilicate Glass: A Multiscale, Multi-technique Approach.** *Target journal: Acta Materialia*.
3. Revelo R.J., Souza M.L., Pena R.B., Yoshida S. & Zanotto E.D. (2025). **Early and advanced crystallization stages in lithia-alumina-silica glasses nucleated by ZrO₂.** *Ceramics International*, Vol. 51(20), pp. 30416–30428. <https://doi.org/10.1016/j.ceramint.2025.01.001>
4. Revelo R.J., Ferreira E.B. & Galeano L.A. (2025). **Glass-ceramics as photocatalysts in water treatment: A review.** *Journal of Environmental Chemical Engineering*, Vol. 13(4), 117245. <https://doi.org/10.1016/j.jece.2025.117245>
5. Barbosa T.V., Dernowsek J.A., Tobar R.J.R., Casali B.C., Fortulan C.A., Ferreira E.B., Selistre-de-Araújo H.S. & Branciforti M.C. (2022). **Fabrication, morphological, mechanical and biological performance of 3D printed poly(ε-caprolactone)/bioglass composite scaffolds for bone tissue engineering applications.** *Biomedical Materials*, Vol. 17(5), 055014. <https://doi.org/10.1088/1748-605X/ac88ad>
6. Fernandes R.G., Reis R.M.C.V., Tobar R.R., Zanotto E.D. & Ferreira E.B. (2019). **Simulation and experimental study of the particle size distribution and pore effect on the crystallization of glass powders.** *Acta Materialia*, Vol. 175, pp. 130–139. <https://doi.org/10.1016/j.actamat.2019.05.049>
7. Revelo R.J., Menegazzo A.P. & Ferreira E.B. (2018). **Cathode-Ray Tube panel glass replaces frit in transparent glazes for ceramic tiles.** *Ceramics International*, Vol. 44(12), pp. 13790–13796. <https://doi.org/10.1016/j.ceramint.2018.04.222>

PRESENTATIONS & INVITED TALKS

- 65th Brazilian Ceramics Congress, 2022. Panelist in "Management, Logistics, and Destination: Circular Economy of Electronic Waste"
- 7th Latin American Congress of Photocatalysis, Photoelectrochemistry, and Photobiology, 2021. Presentation: "Hematite and Magnetite modified glass ceramics as active visible light driven photocatalytic and photo Fenton degradation of water contaminants"
- 12th Colombian Symposium on Catalysis, 2021. Presentation: "Evaluation of the $\text{Bi}_2\text{O}_3\cdot\text{Fe}_2\text{O}_3$ Glass-Ceramic as a Fenton-like Photocatalyst for Degradation of Ibuprofen in Water Decontamination"
- X BrasGlass - Brazilian Symposium on Glass and Related Materials, 2014. Panel: "Reformulation of Ceramic Glazes from Recycled CRT Glass"

PROFESSIONAL DEVELOPMENT

- Disruptive Innovation & Exponential Technologies
- Artificial Intelligence for Everyone — DeepLearning.AI
- Technology Roadmapping
- Pedagogical Practices, Scientific Communication & Oral Presentation

REFERENCES

References include:

- Prof. Dr. Edgar D. Zanotto — Federal University of São Carlos (UFSCar), Brazil (Postdoctoral supervisor)
- Prof. Dr. Eduardo Bellini Ferreira — University of São Paulo (USP), Brazil (Ph.D. & M.Sc. supervisor)
- Prof. Dr. Luis Alejandro Galeano — University of Nariño, Colombia (Postdoctoral supervisor)